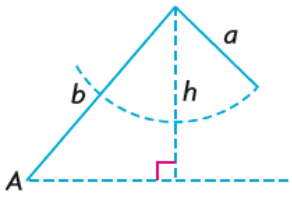
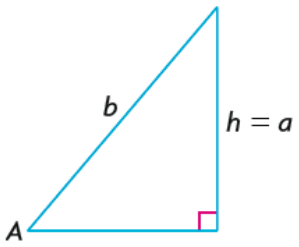
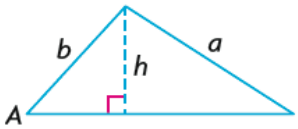
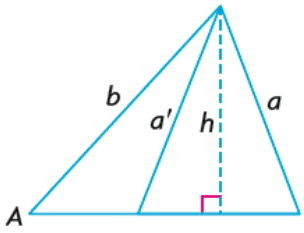
L6 – Ambiguous Case of Sine

MCR3U

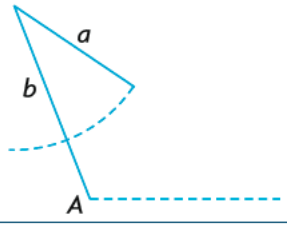
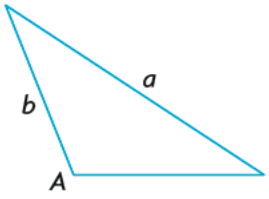
Jensen

If you are given _____ in a triangle, there are many scenarios to consider.

If $\angle A$, a, and b are given and $\angle A$ is acute, there are 4 scenarios to consider:

If $\angle A$ is acute and $a < h$, no triangle exists. 	If $\angle A$ is acute and $a = h$, one right triangle exists. 
If $\angle A$ is acute and $a > b$, one triangle exists. 	If $\angle A$ is acute and $h < a < b$, two triangles exist. 

If $\angle A$, a, and b are given and $\angle A$ is obtuse, there are 2 scenarios to consider:

If $\angle A$ is obtuse and $a < b$ or $a = b$, no triangle exists. 	If $\angle A$ is obtuse and $a > b$, one triangle exists. 

The ambiguous case only occurs when two possible triangles exist for the same given information. This means, the ambiguous case must be considered if _____.

Example 1: In triangle ABC, side a = 12 cm, side b = 17 cm, and A = 21°. Find the measure of angle B.

Example 2: In triangle ABC, side a = 8 cm, side c = 10 cm, and A = 34°. Find angle C.

Example 3: In triangle ABC, side a = 14 cm, side b = 17 cm, and A = 54°. Find the measure of all missing sides and angles.